tenglamani hosil qilamiz. Endi bu chiziqning umumiy chiziq ekanligi

$$\begin{cases} x = \cos\varphi, \\ y = \sin\varphi, \quad 0 \le \varphi \le 2\pi \end{cases}$$

tenglamalar bilan aniqlangan aylananing

$$f : M(\varphi) \to (C\sqrt{2\cos^2\varphi}, \varphi)$$

formula yordamida aniqlangan lokal topologik akslantirishdagi obrazi Bernulli lemniskatasi bilan ustma-ust tushishidan kelib chiqadi.

§ 2. Vektor funksiyalar uchun differensial hisob

Bizning kursimizda vektor analiz muhim rol o'ynaydi. Shuning uchun bu paragrafda qisqacha vektor-funksiyalar ustida to'xtalamiz.

Birorta G to'plam berilgan bo'lsin. Agar G to'plamning har bir nuqtasiga aniq bitta vektor mos qo'yilgan bo'lsa, G to'plamda *vektor funksiya* berilgan deyiladi. Bu moslikni $p \to \vec{r}(p)$ ko'rinishda yozamiz.

Vektor-funksiya uchun limit tushunchasi skalyar funksiyalar limiti kabi kiritiladi.

***Ta'rif.* Berilgan $\vec{r}(p)$ vektor-funksiya va o'zgarmas $\bar{a}$ vektor uchun $p \to p_0$ da $|\vec{r}(p) - \bar{a}| \to 0$ munosabat bajarilsa, $\vec{r}(p)$ vektor $p \to p_0$ da $\bar{a}$ limitga ega deyiladi va $\vec{r}(p) \to \bar{a}$ ko'rinishda yoziladi.**

Bu yerda $|\vec{a}| = \sqrt{(\vec{a}, \vec{a})}$ bo'lib, $(\vec{a}, \vec{a})$ esa skalyar ko'paytmadir. Agar fazoda kiritilgan dekart koordinatalar sistemasida

$$\vec{r}(p) = \{\, x(p), y(p), z(p) \,\}, \qquad \vec{a} = \{a_1, a_2, a_3\}$$

bo'lsa, $p \to p_0$ da $\vec{r}(p) \to \vec{a}$ munosabat quyidagi uchta munosabatga ekvivalentdir:

$$x(p) \underset{p \to p_0}{\to} a_1, \; y(p) \underset{p \to p_0}{\to} a_2, \; z(p) \underset{p \to p_0}{\to} a_3.$$

Vektor-funksiya limiti uchun quyidagi teorema o'rinlidir.

Teorema-6. *Berilgan $\vec{r}(p), \vec{\rho}(p)$ -vektor-funksiyalar va $\lambda(p)$ - skalyar funksiya G to'plamda aniqlangan bo'lib, ular uchun $p \to p_0$ da $\vec{r}(p) \to \vec{a}$, $p \to p_0$ da $\vec{\rho}(p) \to \vec{b}$ va $\lim\limits_{p \to p_0} \lambda(p) = \lambda_0$ tengliklar bajarilsa, quyidagi munosabatlar o'rinlidir;*

$$1). \quad \vec{r}(p) \pm \vec{\rho}(p) \underset{p \to p_0}{\to} \vec{a} \pm \vec{b},$$

$$2). \quad \lambda(p)\vec{r}(p) \underset{p \to p_0}{\to} \lambda_0 \vec{a},$$

$$3). \quad (\vec{r}(p), \vec{\rho}(p)) \underset{p \to p_0}{\to} (\vec{a}, \vec{b})$$

$$4). \quad [\vec{r}(p), \vec{\rho}(p)] \underset{p \to p_0}{\to} [\vec{a}, \vec{b}]$$

Bu yerda $[,]$ – vektor ko'paytma belgisi.

Isbot.

1). Vektor funksiya limiti ta'rifga ko'ra,

$$\vec{d}(p) = \vec{r}(p) \pm \vec{\rho}(p), \quad \vec{c} = \vec{a} \pm \vec{b}$$

belgilashlar kiritib

$$\left|\vec{d}(p) - \vec{c}\right| \underset{p \to p_0}{\to} 0 \qquad (*)$$

munosabatni isbotlaymiz. Buning uchun

$$\left|\vec{d}(p) - \vec{c}\right| = \left|(\vec{r}(p) - \vec{a}) \pm (\vec{\rho}(p) - \vec{b})\right| \le \left|\vec{r}(p) - \vec{a}\right| + \left|\vec{\rho}(p) - \vec{b}\right|$$

tengsizlikni yozib olamiz. Bu tengsizlikning o'ng tomonidagi ifoda $p \to p_0$ da nolga intiladi. Shuning uchun $(*)$ munosabat o'rinli bo'ladi.

2). Ikkinchi munosabat

$$|\lambda(p)\vec{r}(p)-\lambda_0\vec{a}|=|\lambda(p)\vec{r}(p)-\lambda(p)\vec{a}+\lambda(p)\vec{a}-\lambda_0\vec{a}|\le$$

$$\le|\lambda(p)||\vec{r}(p)-\vec{a}|+|\lambda(p)-\lambda_0||\vec{a}|.$$

tengsizlikdan kelib chiqadi.

3). Teoremaning uchinchi tasdig'ini isbotlash uchun skalyar ko'paytmalarni vektorning dekart koordinatalari orqali ifodalash yetarli.

4). Siz analitik geometriya kursidan bilasizki,

Agar,

$$\vec{r}(p)=\{x_1(p),y_1(p),z_1(p)\},\quad \vec{\rho}(p)=\{x_2(p),y_2(p),z_2(p)\}$$

bo'lsa,

$$[\vec{r}(p),\vec{\rho}(p)]=\left\{\begin{vmatrix} y_1(p) & z_1(p)\\ y_2(p) & z_2(p)\end{vmatrix},\begin{vmatrix} z_1(p) & x_1(p)\\ z_2(p) & x_2(p)\end{vmatrix},\begin{vmatrix} x_1(p) & y_1(p)\\ x_2(p) & p_2(p)\end{vmatrix}\right\}$$

tenglik o'rinli bo'ladi. Vektor-funksiya limiti ta'rifiga asosan

$p\to p_0$ da $\quad \vec{r}(p)\to\vec{a},\quad \vec{\rho}(p)\to\vec{b}\quad$ bo'lgani uchun

$$x_1(p)\to a_1,\quad y_1(p)\to a_2,\quad z_1(p)\to a_3 \text{ va}$$

$$x_2(p)\to b_1,\quad y_2(p)\to b_2,\quad z_2(p)\to b_3.$$

munosabatlar o'rinli bo'ladi.

Bu yerda $\quad \vec{a}=\{a_1,a_2,a_3\},\quad \vec{b}=\{b_1,b_2,b_3\}.$

Shuning uchun,

$$[\vec{r}(p),\vec{\rho}(p)]\underset{p\to p_0}{\to}\left\{\begin{vmatrix} a_2 & a_3\\ b_2 & b_3\end{vmatrix},\begin{vmatrix} a_3 & a_1\\ b_3 & b_1\end{vmatrix},\begin{vmatrix} a_1 & a_2\\ b_1 & b_2\end{vmatrix}\right\}$$

munosabatni hosil qilamiz.

Vektor funksiyalar uchun uzluksizlik va differensiallanuvchanlik tushunchalari skalyar funksiyalar uzluksizligi va diffrensiallanuvchiligi tushunchalari kabi kiritiladi.

Agar $\vec{r}(p) \underset{p \to p_0}{\longrightarrow} \vec{r}(p_0)$ munosabat bajarilsa, $\vec{r}(p)$ vektor-funksiya p_0 nuqtada *uzluksiz* deyiladi. Limit ta'rifiga ko'ra $\vec{r}(p)$ vektor-funktsiyaning p_0 nuqtada uzluksizligi $x(p), y(p), \ z(p)$ funksiyalarning p_0 nuqtada uzluksizligiga ekvivalentdir.

Teorema-7. *Berilgan $\vec{r}(p)$ va $\vec{\rho}(p)$ − vektor funksiyalar va $\lambda(p)$ funksiya p_0 nuqtada uzluksiz bo'lsa, $\lambda(p)\cdot\vec{r}(p)$, $\vec{r}(p)\pm\vec{\rho}(p)$, $[\vec{r}(p),\vec{\rho}(p)]$ vektor funksiyalar va $(\vec{r}(p),\vec{\rho}(p))$ funksiya ham p_0 nuqtada uzluksizdir.*

Bu teorema 1-teoremadan bevosita kelib chiqadi.

Endi hosila tushunchasini kiritamiz. Vektor funksiya aniqlangan G to'plam sonlar o'qining qism to'plami bo'lsa, $\vec{r}(p)$ vektor-funksiya bir o'zgaruvchili vektor-funksiya bo'ladi. Agar $p_0 \in G$ nuqta uchun

$$\lim_{h \to 0} \frac{\vec{r}(p_0+h)-\vec{r}(p_0)}{h}$$

mavjud bo'lsa, uni $\vec{r}'(p_0)$ bilan belgilaymiz va $\vec{r}(p)$ vektor-funksiyaning p_0 nuqtadagi *hosilasi* deb ataymiz.

Agar

$$\lim_{h \to 0} \frac{\vec{r}(p_0+h)-\vec{r}(p_0)}{h} = \vec{r}'(p_0)$$

tenglikni koordinatalar orqali yozsak u

$$\vec{r}'(p_0)=\{x'(p_0), \quad y'(p_0), \quad z'(p_0)\}$$

ko'rinishda bo'ladi. Demak, $\vec{r}(p)$ − vektor-funksiya p_0 nuqtada hosilaga ega bo'lishi uchun $\{x'(p_0), \ y'(p_0), \ z'(p_0)\}$ hosilalarning mavjud bo'lishi zarur va yetarlidir.

Agar G to'plam tekislikdagi birorta soha bo'lsa, $p=(u,v)$ belgilash kiritamiz. Bu holda $\vec{r}(p)$ va uning koordinata funksiyalari $x(u,v), y(u,v), z(u,v)$ ikki o'zgaruvchili funksiyalar bo'ladi. Demak, endi xususiy hosilalar haqida gapirishimiz mumkin.

Yuqoridagi hosila tushunchasidan foydalanib,

$$\vec{r}'_u(u,v)=\{x'_u,\quad y'_u,\quad z'_u\} \text{ va } \vec{r}'_v(u,v)=\{x'_v,\quad y'_v,\quad z'_v\}$$

tengliklarni hosil qilamiz.

Hosilalar uchun quyidagi teoremani isbotlaymiz.

Teorema-8. $\vec{r}(p)$, $\vec{\rho}(p)$ *−vektor-funksiyalar va* $\lambda(p)$ *funksiya* p_0 *nuqtada differensiallanuvchi bo'lsa, quyidagi munosabatlar o'rinlidir:*

$$1).\quad (\lambda(p)\vec{r}(p))'=\lambda' r+\lambda\vec{r}',$$

$$2).\quad (\vec{r}(p)\pm\vec{\rho}(p))'=\vec{r}'(p)\pm\vec{\rho}'(p),$$

$$3).\quad (\vec{r}(p),\vec{\rho}(p))'=(\vec{r}'(p),\vec{\rho}(p))+(\vec{r}(p),\vec{\rho}'(p)),$$

$$4).\quad [\vec{r}(p),\vec{\rho}(p)]'=[\vec{r}'(p),\vec{\rho}(p)]+[\vec{r}(p),\vec{\rho}')].$$

Isbot.

1). Berilgan $\vec{r}(p)$ vektor-funksiyani $\lambda(p)$ ga ko'paytirib

$$\lambda(p)\vec{r}(p)=\{\lambda(p)x(p),\quad \lambda(p)y(p),\quad \lambda(p)z(p)\}$$

tenglikni yozsak, darhol bu tenglikdan

$$(\lambda(p)\vec{r}(p))'=\lambda'(p)\vec{r}(p)+\lambda(p)\vec{r}'$$

munosabat kelib chiqadi.

2). Ikkinchi tenglik isbotini o'quvchilarga qoldirib, 3-va 4-tengliklarni isbotlaylik. Uchinchi tenglikda skalyar ko'paytma skalyar miqdor bo'lganligi uchun uning hosilasini topish uchun

$$\lim_{h\to 0}\frac{(\vec{r}(p+h),\vec{\rho}(p+h))-(\vec{r}(p),\vec{\rho}(p))}{h}$$

limitni hisoblaymiz.

Buning uchun

$$(\vec{r}(p+h),\vec{\rho}(p+h))-(\vec{r}(p),\vec{\rho}(p))=(\vec{r}(p+h)-\vec{r}(p),\vec{\rho}(p+h))+$$
$$+(\vec{r}(p),\vec{\rho}(p+h)-\vec{\rho}(p))$$

tenglikdan foydalanib,

$$(\vec{r}(p),\vec{\rho}(p))'=\lim_{h\to 0}(\frac{\vec{r}(p+h)-\vec{r}(p)}{h},\vec{\rho}(p+h))+$$

$$+\lim_{h\to 0}(\vec{r}(p),\frac{\vec{\rho}(p+h)-\vec{\rho}(p)}{h}).$$

tenglikni hosil qilamiz. Bu tenglikning o'ng tomoni

$$(r'(p), \vec{\rho}(p)) + (\vec{r}(p), \vec{\rho}'(p))$$

miqdorga tengdir.

4-tenglikni isbotlash uchun

$$[\vec{r}(p), \vec{\rho}(p)]' = \lim_{h\to 0} \frac{[\vec{r}(p+h), \vec{\rho}(p+h) - [\vec{r}(p), \vec{\rho}(p)]}{h}$$

tenglikda

$$[\vec{r}(p+h), \vec{\rho}(p+h)] - [\vec{r}(p), \vec{\rho}(p)] = [\vec{r}(p+h), \vec{\rho}(p+h)] -$$

$$-[\vec{r}(p), \vec{\rho}(p+h)] + [\vec{r}(p), \vec{\rho}(p+h)] - [\vec{r}(p), \vec{\rho}(p)]$$

munosabatdan foydalanish yetarlidir.□

Endi bir o'zgaruvchili $\vec{r}(t)$ vektor-funksiya uchun integral tushunchasini kiritaylik. Agar $\vec{r}(t)$ vektor-funktsiya uchun diffrensiallanuvchi $\vec{\rho}(t)$ vektor-funksiya mavjud bo'lib, $\vec{r}(t) = \vec{\rho}'(t)$ tenglik bajarilsa, $\vec{\rho}(t)$ vektor-funksiya $\vec{r}(t)$ vektor-funksiyaning ***aniqmas integrali*** deyiladi va quyidagi ko'rinishda yoziladi.

$$\vec{\rho}(t) = \int \vec{r}(t)dt$$

Aniq integral esa quyidagi formula yordamida aniqlanadi.

$$\int_a^b \vec{r}(t)dt = \vec{\rho}(b) - \vec{\rho}(a),$$

Vektor-funksiyaning integrallari uchun quyidagi formulalar bevosita integral va hosila ta'riflari yordamida isbotlanadi:

$$\int_a^b \vec{r}(t)dt = \int_a^c \vec{r}(t)dt + \int_c^b \vec{r}(t)dt,$$ bu yerda, $a \le c \le b$.

$$\int_a^b \lambda \cdot \vec{r}(t)dt = \lambda \cdot \int_a^b \vec{r}(t)dt,$$ bu yerda, $\lambda -$ o'zgarmas son.

$$\int_a^b (\vec{r}, \vec{\rho}(t))dt = (\vec{r}, \int_a^b \vec{\rho}(t)dt),$$ bu yerda, $\vec{r}$ – o'zgarmas vektor.

$$\int_a^b [\vec{r},\vec{\rho}(t)]dt = [\vec{r},\int_a^b \vec{\rho}(t)dt],$$ bu yerda, $\vec{r}$ – o'zgarmas vektor.

Bu tengliklardan uchinchisini isbotlab, qolganlarini o'quvchilarga havola etamiz.

Buning uchun

$$\mu(t) = (\vec{r},\int \vec{\rho}(t)), \qquad \lambda(t) = \int (\vec{r},\vec{\rho}(t))dt$$

belgilashlar kiritamiz va $\vec{r}$ – o'zgarmas vektor ekanligini hisobga olib, $\mu(t)$, $\lambda(t)$ – skalyar differensiallanuvchi funksiyalarning hosilalarini topamiz.

$\vec{\mu}'(t) = (\vec{r},\vec{\rho}(t))$, $\lambda'(t) = (\vec{r},\vec{\rho}(t))$ ya'ni hosilalari tengdir. Demak,

$$\int_a^b \vec{\mu}'(t)dt = \int_a^b \lambda'(t)dt$$

integrallar ham teng bo'ladi.

Endi differensiallanuvchi vektor-funksiya uchun quyidagi muhim teoremani isbotlaymiz.

Teorema-9. *Biror segmentda aniqlangan* $\vec{r}(t)$ *vektor-funksiya uchun quyidagilar o'rinlidir:*

1). $\vec{r}(t)$ *vektor funksiyaning uzunligi o'zgarmas bo'lishi uchun* $\vec{r}(t) \perp \vec{r}'(t)$ *shartning bajarilishi zarur va yetarlidir.*

2). $\vec{r}(t)$ *vektor-funksiyaning yo'nalishi o'zgarmas bo'lishi uchun,* $\vec{r}(t) // \vec{r}'(t)$ *shartning bajarilishi zarur va yetarlidir.*

Isbot. 1). Uzunlik kvadrati uchun $|\vec{r}(t)|^2 = (\vec{r}(t),\vec{r}(t))$ tenglik o'rinli bo'lganligi uchun, $\frac{d}{dt}|\vec{r}(t)|^2 = 2(\vec{r}(t),\vec{r}'(t))$ tenglik o'rinli bo'ladi. Demak,

$$|\vec{r}(t)|^2 = const \quad \Leftrightarrow \quad (\vec{r}(t),\vec{r}'(t)) = 0.$$

2). Agar $\vec{r}(t)$ vektor-funksiyaning yo'nalishi o'zgarmas bo'lsa, uni $\vec{r}(t) = \lambda(t)\cdot\vec{e}$ ko'rinishda yozamiz, bu yerda $\vec{e}$ vektor

$\vec{r}(t)$ yo'nalishdagi birlik vektordir.Yuqoridagi tenglikni differensiallab $\vec{r}'(t)=\lambda'(t)\cdot\vec{e}$ tenglikni hosil qilamiz. Demak $\vec{r}$ va $\vec{r}'$ vektorlar kollineardir.

Agar $\vec{r}$ va $\vec{r}'$ kollinear bo'lsa, ya'ni birorta $\lambda(t)$ funksiya uchun $\vec{r}'(t)=\lambda(t)\cdot\vec{r}(t)$ tenglik o'rinli bo'lsa, $\vec{r}(t)$ vektorning yo'nalishi o'zgarmas ekanligini ko'rsataylik. Buning uchun $\vec{r}(t)=|\vec{r}(t)|\,\vec{e}(t)$ tenglikni yozib, uni differensiallaymiz. Bu yerda $\vec{e}(t)$ birlik vektor va uning yo'nalishi $\vec{r}(t)$ yo'nalishi bilan bir xildir. Agar $\vec{r}(t)=\vec{0}$ bo'lsa, $\vec{e}(t)$ vektor sifatida yo'nalishi ixtiyoriy differensiallanuvchi birlik vektor-funksiyani olish mumkin. Demak, endi

$\vec{r}'(t)=|\vec{r}(t)|'\,\vec{e}'(t)+|\vec{r}(t)|\,\vec{e}'(t)$ tenglikni hosil qilamiz. Bundan esa,

$|\vec{r}(t)|'\,\vec{e}(t)+|\vec{r}(t)|\vec{e}'(t)=\lambda(t)|\vec{r}(t)|\vec{e}(t)$ yoki

$\left(|\vec{r}(t)|'-\lambda(t)|\vec{r}(t)|\right)\vec{e}(t)+|\vec{r}(t)|\vec{e}'(t)=\vec{0}$ tenglik kelib chiqadi.

Bu yerda $\vec{e}(t)$ va $\vec{e}'(t)$ o'zaro perpendikulyar vektorlar bo'lganligi uchun bu tenglikdan $\frac{d\vec{e}(t)}{dt}=\vec{0}$ kelib chiqadi. Demak, $\vec{e}(t)$ - o'zgarmas vektordir. □

Bu paragraf oxirida $n-$marta differensiallanuvchi $\vec{r}(t)$ vektor funksiya uchun Teylor qatorini keltiramiz. Vektor funksiya uchun Teylor qatorini yozish uchun fazoda dekart koordinatalar sistemasini aniqlovchi ortonormal $\vec{i}$, $\vec{j}$, $\vec{k}$ bazisda

$$\vec{r}(t)=x(t)\vec{i}+y(t)\vec{j}+z(t)\vec{k}$$

tenglikni yozib, $x(t),y(t),z(t)$ funksiyalar uchun Teylor qatorini yozamiz:

$$x(t+\Delta t)=x(t)+x'(t)\Delta t+x''(t)\frac{\Delta t^2}{2!}+...+x^{(n)}(t)\frac{\Delta t^n}{n!}+\frac{\Delta t^n}{n!}\varepsilon_1(t,\Delta t)$$

$$y(t+\Delta t)=y(t)+y'(t)\Delta t+y''(t)\frac{\Delta t^2}{2!}+...+y^{(n)}(t)\frac{\Delta t^n}{n!}+\frac{\Delta t^n}{n!}\varepsilon_2(t,\Delta t)$$

$$z(t+\Delta t)=z(t)+z'(t)\Delta t+z''(t)\frac{\Delta t^2}{2!}+...+z^{(n)}(t)\frac{\Delta t^n}{n!}+\frac{\Delta t^n}{n!}\varepsilon_3(t,\Delta t)$$

Bu tengliklardan

$$\vec{r}(t+\Delta t)=\vec{r}(t)+\vec{r}\,'(t)\Delta t+\vec{r}\,''(t)\frac{\Delta t^2}{2!}+...+\vec{r}^{(n)}(t)\frac{\Delta t^n}{n!}+\frac{\Delta t^n}{n!}\vec{\varepsilon}(t,\Delta t)$$

qatorni hosil qilamiz. Bu yerda

$$\vec{\varepsilon}(t,\Delta t)=\{\varepsilon_1(t,\Delta t),\varepsilon_2(t,\Delta t),\varepsilon_3(t,\Delta t)\}$$

va $\left|\vec{\varepsilon}(t,\Delta t)\right|\underset{\Delta t\to 0}{\longrightarrow}0$ munosabat o'rinlidir.

Masalan, $n=1$ va $n=2$ bo'lganda

$$\vec{r}(t+\Delta t)=\vec{r}(t)+\vec{r}\,'(t)\Delta t+\Delta t\vec{\varepsilon}(t,\Delta t)$$

$$\vec{r}(t+\Delta t)=\vec{r}(t)+\vec{r}\,'(t)\Delta t+\vec{r}\,''(t)\frac{\Delta t^2}{2!}+\frac{\Delta t^2}{2!}\vec{\varepsilon}(t,\Delta t)$$

qatorlar hosil bo'ladi.

Mashqlar va masalalar

1. Birorta $[a,b]$ kesmada aniqlangan $\vec{r}(t)$ vektor funksiya uchun $\vec{r}(t)\neq\vec{0}$, $\vec{r}\,'(t)\neq\vec{0}$ va $\vec{r}(t)\parallel\vec{r}\,'$ shartlar har bir $t\in[a,b]$ uchun bajarilsa,

$$\vec{r}=\vec{r}(t)$$

tenglama to'g'ri chiziq kesmasini aniqlashini ko'rsataylik.

Buning uchun $\vec{r}(t)=\lambda(t)\vec{r}\,'(t)$ tenglikni yozib $\vec{e}(t)=\dfrac{\vec{r}(t)}{|\vec{r}(t)|}$ vektorning o'zgarmas vektor ekanligini ko'rsataylik. Hosilani hisoblab

$$\frac{d}{dt}\vec{e}(t)=\frac{|\vec{r}(t)\vec{r}'(t)|-\vec{r}(t)\dfrac{(\vec{r}(t),\vec{r}'(t))}{|\vec{r}(t)|}}{\vec{r}^{\,2}(t)}=\frac{\lambda^2|\vec{r}'(t)|^2\vec{r}'(t)-\lambda^2|\vec{r}'(t)|^2\vec{r}'(t)}{|\vec{r}(t)|^3}=\vec{0}$$

tenglikni olamiz. Demak $\vec{r}(t)$ yo'nalishi o'zgarmas vektor va uning uchi to'g'ri chiziq kesmasini chizadi.

2. Tekislikdagi birorta G sohada aniqlangan differensiallanuvchi $\vec{r}(u,v)$ vektor-funksiya berilgan. Berilgan $\vec{r}(u,v)$ vektor-funksiyaning uzunligi o'zgarmas bo'lishi uchun, $(\vec{r},\vec{r}_u)=(\vec{r},\vec{r}_v)=0$ tengliklarning bajarilishi zarur va yetarlidir. Bu faktni isbotlash uchun

$$|\vec{r}|^2=(\vec{r},\vec{r})$$

tenglikdan foydalanamiz. Agar $|\vec{r}(u,v)|=const$ bo'lsa,

$$0=\frac{d}{du}|\vec{r}|^2=\frac{d}{du}(\vec{r},\vec{r})=2(r,\vec{r}_u)$$

$$0=\frac{d}{du}|\vec{r}|^2=\frac{d}{du}(\vec{r},\vec{r})=2(r,\vec{r}_v)$$

tengliklardan $(\vec{r},\vec{r}_u)=(\vec{r},\vec{r}_v)=0$ tengliklar kelib chiqadi.

Endi $\vec{r}\perp\vec{r}_u$, $\vec{r}\perp\vec{r}_v$ bo'lsin deb faraz qilaylik. Bu holda

$$\frac{d}{du}|\vec{r}|^2=2(r,\vec{r}_u)=0,\ \frac{d}{du}|\vec{r}|^2=2(r,\vec{r}_v)=0$$

tengliklardan $|\vec{r}(u,v)|$ funksiyaning o'zgarmas ekanligi kelib chiqadi.

3. Tekislikdagi birorta G sohada aniqlangan differensiallanuvchi $\vec{r}(u,v)$ vektor-funksiyaning $\vec{r}_u$, r_v vektorlarning ikkalasiga ham kolleniar bo'lishi uning yo'nalishi o'zgarmas ekanligiga teng kuchli ekanligini ko'rsataylik.

Agar $\vec r(u,v)$ vektor-funksiyaning yo'nalishi o'zgarmas bo'lsa, uni $\vec r(u,v)=\lambda(u,v)\vec e$ ko'rinishda yozish mumkin. Bu yerda $\lambda(u,v)-$skalyar funksiya bo'lib, $\vec e-$o'zgarmas birlik vektordir. Bu ko'rinishdan $\vec r_u=\lambda_u(u,v)\vec e,\quad \vec r_v=\lambda_v(u,v)\vec e$ tengliklarni hosil qilamiz. Demak $\vec r(u,v)$ vektor $\vec r_u$, $\vec r_v$ vektorning ikkalasiga ham kollineardir.

Endi $\vec r(u,v)=\lambda(u,v)\vec r_u$, $\vec r(u,v)=\lambda(u,v)\vec r_v$, tengliklar o'rinli deb faraz qilib, $\vec e(u,v)=\dfrac{\vec r(u,v)}{|\vec r(u,v)|}$ vektorning o'zgarmas vektor ekanligini ko'rsataylik. Buning uchun $\dfrac{d}{du}\vec e=\vec 0,\quad \dfrac{d}{dv}\vec e=\vec 0,$ tengliklarni isbotlaymiz.

$$\frac{d}{du}\vec e=\frac{\vec r_u|\vec r|-\vec r\,\dfrac{(\vec r,\vec r_u)}{|\vec r|}}{|\vec r^{\,2}|}=\frac{\vec r_u|\vec r|^2-\vec r(\vec r,\vec r_u)}{|\vec r|^3}=\frac{\lambda^2|\vec r_u|^2\vec r_u-\lambda^2|\vec r_u|^2\vec r_u}{|\vec r|^3}=\vec 0$$

Xuddi shunday

$$\frac{d}{dv}\vec e=\frac{\mu^2|\vec r_v|^2\vec r_v-\mu^2|\vec r_v|^2\vec r_v}{|\vec r|^3}=\vec 0$$

tenglikni olamiz. Demak $\vec r(u,v)=|\vec r(u,v)|\vec e$ bo'lib, $\vec r$ vektorning yo'nalishi o'zgarmasdir.

4. Birorta G sohada aniqlangan differensiallanuvchi $\vec r(u,v)$ vektor-funksiyaning xususiy hosilalari $\vec r_u$, $\vec r_v$ vektorlar nol vektor bo'lishi $\vec r(u,v)$ ning o'zgarmas vektor bo'lishiga teng kuchli ekanligini ko'rsataylik.

Xususiy hosilalar uchun

$$\vec r_u=\vec 0,\ \vec r_v=\vec 0$$

tengliklar o'rinli bo'lsa, $\vec r(u,v)$ ning koordinata funksiyalari uchun

$$x_u = 0, \quad x_v = 0$$
$$y_u = 0, \quad y_v = 0$$
$$z_u = 0, \quad z_v = 0$$

tengliklarni hosil qilamiz. Demak $x(u,v), y(u,v), z(u,v)$ funksiyalar o'zgarmasdir. Bundan esa

$$\vec{r}(u,v) = \{x(u,v), y(u,v), z(u,v)\}$$

vektorning o'zgarmas ekanligi kelib chiqadi. Aksincha, $\vec{r}(u,v)$ vektor-funksiyaning o'zgarmas vektor ekanligidan $x(u,v), y(u,v), z(u,v)$ funksiyalar o'zgarmas bo'lishi kelib chiqadi. Bundan esa

$$\vec{r}_u = \vec{0}, \ \vec{r}_v = \vec{0}$$

tengliklar hosil bo'ladi.

§ 3. Egri chiziq urinmasi va normal tekisligi

Elementar γ egri chiziqning M nuqtasidan o'tuvchi urinma tushunchasini kiritib, uning tenglamasini keltirib chiqaraylik. Buning uchun M nuqtadan l to'g'ri chiziqni o'tkazaylik, N bilan M ga yaqin bo'lgan γ chiziqning birorta nuqtasini belgilaylik. Egri chiziqdagi M va N nuqtalar orasidagi masofani d bilan, N nuqtadan l- to'g'ri chiziqqacha bo'lgan masofani h bilan belgilaylik. Agar, N nuqta M ga yaqinlasha borsa, tabiiyki, d va h masofalar nolga intiladi. Lekin, $\frac{h}{d}$ ifodaning nimaga intilishi haqida hech narsa deya olmaymiz.

<u>**Ta'rif**</u>: *Egri chiziq γ ning N nuqtasi M ga intilganda $\frac{h}{d}$ ifoda nolga intilsa, l-to'g'ri chiziq, γ ning M nuqtadagi urinmasi deb ataladi.*